

- 6 (a) (i) When the liquid boils, the molecules in the liquid are totally separated with a large increase in the inter-molecular distance. This results in a significant increase in the potential energy of the molecules. [B1]

The kinetic energy of the molecules does not change as the temperature remains constant during boiling. [B1]

The internal energy of a system is the sum of a random distribution of kinetic and potential energies associated with the molecules of a system. Hence, the internal energy of the liquid increases due to the increase in the potential energy of the molecules in the liquid. [B1]

$$\begin{aligned}
 \text{(ii) Heat loss} &= Q_{\text{steam convert to water at } 100^{\circ}\text{C}} + Q_{\text{water cooled to } 50^{\circ}\text{C}} \\
 &= m_s L_v + m_s c_w \Delta T_{100 \rightarrow 50^{\circ}\text{C}} \\
 &= m_s (2.3 \times 10^6) + m_s (4.2 \times 10^3)(100 - 50) \text{ [C1]}
 \end{aligned}$$

$$\begin{aligned}
 \text{Heat gain} &= Q_{\text{glass heated to } 50^{\circ}\text{C}} + Q_{\text{water heated to } 50^{\circ}\text{C}} \\
 &= m_g c_g \Delta T_{20 \rightarrow 50^{\circ}\text{C}} + m_w c_w \Delta T_{20 \rightarrow 50^{\circ}\text{C}} \\
 &= (0.100)(8.4 \times 10^2)(50 - 20) + (0.200)(4.2 \times 10^3)(50 - 20) \text{ [C1]}
 \end{aligned}$$

$$\begin{aligned}
 \text{Heat loss} &= \text{Heat gain} \\
 m_s &= 0.0110 \text{ kg} \quad \text{[A1]}
 \end{aligned}$$

- (iii) The specific latent heat of vaporisation of steam is $2.3 \times 10^6 \text{ J kg}^{-1}$. This means that 1 kg of 100°C steam will release $2.3 \times 10^6 \text{ J}$ of thermal energy as it condenses to 100°C water. And after it releases this much of energy onto the skin, it is now identical to 100°C water and will continue to burn the skin.

Hence 100°C steam will cause a more serious burn as it releases more / extra amount of thermal energy as compared to 100°C water. [A1] [M1]

or
When there is conversion of phase (gaseous to liquid), extra energy will be release.

or
For 1 kg of steam, 10^6 J of energy will be released.
For 1 kg of 100°C water, at most 10^5 J of energy will be released.
[no need to give values]

- (b) (i) An ideal gas is one which obeys the ideal gas equation ($pV = nRT$) at all temperatures, volumes and pressures. [B1] [B1]

- (ii) When the gas is being compressed, the piston moves to the right and it collides with the gas molecules, the molecules will rebound with a higher speed and thus average kinetic energy increases. [B1]



Since average kinetic energy is proportional to temperature, [B1]
temperature increases.

(iii) 1. $pV = nRT$

$$n = \frac{pV}{RT}$$

$$= \frac{1.1 \times 10^5 (6.2 \times 10^{-4})}{8.31(28 + 273)} \quad [C1]$$

$$= 0.0273 \text{ mol} \quad [A1]$$

2. $Q = 0 \text{ J}$ (as process happens quickly, no time for heat exchange) [M1]

$$\Delta U = Q + W_{on}$$

$$\Delta U = 0 + 15 \quad [A1]$$

$$= 15 \text{ J}$$

3. $N = nN_A$

$$= 0.0273(6.02 \times 10^{23})$$

$$= 1.64 \times 10^{22} \text{ molecules} \quad [M1]$$

$$U = KE \quad (\text{for ideal gas, } PE = 0)$$

$$\Delta KE_{\text{total}} = \Delta U_{\text{total}}$$

$$\Delta KE_{\text{one molecule}} = \frac{\Delta U_{\text{total}}}{N}$$

$$= \frac{15}{1.64 \times 10^{22}}$$

$$= 9.15 \times 10^{-22} \text{ J} \quad [A1]$$

(c) (iv) When the gas is compressed slowly, the temperature T remains constant.
For the process that is compressed quickly, the temperature increases.
The final temperature of the gas that compressed slowly is lower than the final temperature of the gas that compressed quickly (in (b)). [M1]

Using the ideal gas equation, $pV = nRT$, with the same final volume (n & R remaining constant), the final pressure of the gas that is compressed slowly is lower than the gas that is compressed quickly. [A1]
(or the final pressure of the gas is greater for the gas that is compressed quickly).

